

Consider a quadratic function

$$f = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{g}^T \mathbf{x} + c,$$

where $\mathbf{Q} = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$, $\mathbf{g} \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

1. For which values of α f does not have a local minimum?
2. For which values of α f has a unique global minimum?
3. What are the conditions on $\mathbf{g}$ and α so that the problem has an infinite number of global minima?